
Microsoft 1 Cr+ Roadmap (Java Focus)

Alright 🚀 You're aiming for a Microsoft + 1 Cr+ package (SDE2+ / Research roles / Top-tier Global offers). That's ambitious — but totally possible if you follow a structured multi-year roadmap. Since you want to use Java (not C++), I'll create a full roadmap from your second year → final year → placements/internships.

Phase 1: Java + CS Foundations (Now – 6 months)

Goal: Be fluent in Java and core programming.

- ☒ **Java Basics:** variables, loops, functions, arrays.
- ☒ **OOP in Java:** classes, objects, inheritance, polymorphism, abstraction, encapsulation.
- ☒ **Exception Handling, File Handling, Multithreading.**
- ☒ **Collections Framework:** (ArrayList, HashMap, HashSet, Queue, Stack, TreeMap).
- ☒ **Generics, Streams API, Lambda expressions.**
- ☒ **Big O Notation basics.**

Resources:

- **Book:** *Head First Java*
 - **Playlist:** Kunal Kushwaha Java + DSA
 - **LeetCode:** Easy → Medium, 100 problems in 6 months.
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Phase 2: Data Structures & Algorithms (1 year)

Goal: Master problem-solving, which is core for FAANG/Microsoft interviews.

- **Arrays & Strings** (sliding window, two-pointer)
- **Recursion & Backtracking**
- **Linked List, Stack, Queue**
- **Binary Search + Advanced Binary Search**
- **Hashing & Maps**
- **Trees** (BST, Binary Trees, Trie, Segment Trees)
- **Graphs** (BFS, DFS, Dijkstra, Union-Find, Topological Sort)
- **Dynamic Programming** (all patterns: knapsack, LIS, matrix DP)

Resources:

- **DSA in Java:** Aditya Verma (YouTube, DP + Heap)
 - **Striver's SDE Sheet:** (191 problems)
 - **LeetCode:** 500–700 problems in 1.5 years.
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Phase 3: Core CS Subjects (Parallel with DSA)

Goal: Crack system design + low-level interview rounds.

- ☒ **Operating System:** (Processes, Threads, Scheduling, Deadlocks, Memory Mgmt)
- ☒ **DBMS:** (Normalization, Indexing, Transactions, SQL queries)
- ☒ **Computer Networks:** (OSI, TCP/IP, HTTP/HTTPS, DNS, Routing)
- ☒ **OOP:** (in Java, with design principles SOLID)

Resources:

- **OS:** *Galvin (Silberschatz)*
 - **DBMS:** Navathe / GFG + SQLZoo
 - **CN:** *Kurose & Ross*
 - **OOP + Design:** *Head First Design Patterns*
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Phase 4: Projects + Development (2nd → 3rd Year)

Goal: Stand out with strong projects + open source.

- **Java-based projects:**
 - Banking/Finance System with Java + MySQL
 - Spring Boot REST APIs (Microservices)
 - Chat Application (JavaFX + Sockets)
- **Web/Cloud:**
 - Full-stack project (React + Spring Boot + MongoDB)
 - Deploy on AWS/Azure
- **Open Source:** Contribute to Apache / Spring.

Resources:

- Spring Boot docs
 - System Design Primer (GitHub)
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Phase 5: System Design + Advanced (3rd Year → Internships)

Goal: Crack high package roles (SDE2+).

- **Low-Level Design (LLD):** Classes, relationships, SOLID, UML diagrams.
- **High-Level Design (HLD):** Scalability, load balancers, caching, sharding, databases.
- **Case Studies:** Design Netflix, WhatsApp, Uber.

Resources:

- *Grokking the System Design Interview*
 - *Designing Data-Intensive Applications (DDIA)*
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Phase 6: Interview Prep (Final Year)

-  **Resume:** 2 big projects + internships + DSA stats (700+ problems).
 -  **Mock Interviews:** Pramp, InterviewBit, LeetCode contests.
 -  **Company-specific prep:** Microsoft interview archives (GFG, LeetCode Discuss).
 -  **Competitive Coding:** (Codeforces, AtCoder → improves speed).
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Minimum Requirements for 1 Cr+ Package

- **DSA:** 600–700 problems (LeetCode + Striver's + Contests).
 - **Core CS:** Strong in OS, DBMS, CN, OOP.
 - **Projects:** 2–3 industry-level, deployed projects (Spring Boot + Cloud).
 - **System Design:** LLD + HLD mastery.
 - **Internship:** Microsoft / Google / Top product company.
 - **Competitive Edge:** ICPC/Codeforces rating helps for global roles.
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Timeline (You in 2nd Year Now)

- **2nd Year (Sem 3–4):** Java + OOP + Basic DSA + Mini Projects
- **3rd Year (Sem 5–6):** Advanced DSA + Core CS + Internships + Big Projects
- **4th Year (Sem 7–8):** System Design + Interview Prep + Global Opportunities

⚡ If you stick to this roadmap with discipline → you'll be interview-ready for Microsoft, Google, Amazon, etc. and competitive for 1 Cr+ global offers.

Here's your **Microsoft 1Cr Roadmap (Java) → Free Resources Pack** 📦 :

📌 Phase 1: Java + CS Foundations

- **Java Basics + OOP**
 - [Kunal Kushwaha – OOP in Java](#)
 - [Kunal Kushwaha – Java + DSA Bootcamp \(Basics included\)](#)
 - [Oracle Java Official Tutorial](#)
 - **Collections Framework**
 - [Telusko – Java Collections Playlist](#)
 - **Big O Notation**
 - [Big O Notation Explained – FreeCodeCamp](#)
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📌 Phase 2: Data Structures & Algorithms (DSA in Java)

- **Full DSA**
 - [Kunal Kushwaha DSA Bootcamp \(Complete in Java\)](#)
 - [Aditya Verma – Dynamic Programming Playlist](#)
 - **Practice Sheets**
 - [Striver's SDE Sheet \(191 problems\)](#)
 - [LeetCode \(free problems\)](#)
 - [InterviewBit – Practice DSA](#)
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Phase 3: Core CS Subjects

- **Operating System (OS)**
 - [Neso Academy – OS Playlist](#)
 - **DBMS + SQL**
 - [Neso Academy – DBMS Playlist](#)
 - [SQLZoo – Interactive SQL Practice](#)
 - [LeetCode SQL Problems](#)
 - **Computer Networks (CN)**
 - [Neso Academy – CN Playlist](#)
 - **OOP + Design Principles**
 - [GeeksforGeeks – OOP Concepts](#)
 - [GitHub – Design Patterns in Java Examples](#)
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Phase 4: Projects + Development

- **Spring Boot + REST APIs**
 - [Spring Boot Official Docs](#)
 - [Amigoscode – Spring Boot Crash Course](#)
 - **Java + MySQL Project (Hands-on)**
 - [Spring Boot + MySQL CRUD Tutorial](#)
 - **Cloud Deployment**
 - [AWS Free Tier Signup](#)
 - [Microsoft Azure – Free Learning Paths](#)
 - **Open Source**
 - [Apache Software Foundation Projects](#)
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Phase 5: System Design (LLD + HLD)

- **Low-Level Design (LLD)**
 - [Machine Coding Round Questions \(GitHub\)](#)
 - **High-Level Design (HLD)**
 - [System Design Primer – GitHub](#)
 - [ByteByteGo – Free System Design Resources](#)
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Phase 6: Interview Prep

- **Mock Interviews**
 - [Pramp – Free Mock Interviews](#)
 - [InterviewBit – Free Mock Tests](#)
- **Competitive Programming**
 - [Codeforces](#)
 - [AtCoder](#)
- **Company-specific Questions**
 - [GFG – Microsoft Interview Questions](#)
 - [LeetCode Discuss – Company Archives](#)

✅ All the above are **100% free** resources.

⚡ This means you can go from **beginner** → **1Cr Microsoft-ready** without spending a single rupee.

<https://chatgpt.com/share/68a76d0e-f928-8005-a011-ebd6a1540679>